Competitive Zero Forcing

Paths and Cycles

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Texas Undergraduate Mathematics Conference

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with support from NSF and the American Institute of Mathematics
NSF Grant #2331634

Outline

- Background and ordinary zero forcing
- Competition and our research questions
- Results on paths
- Results on cycles

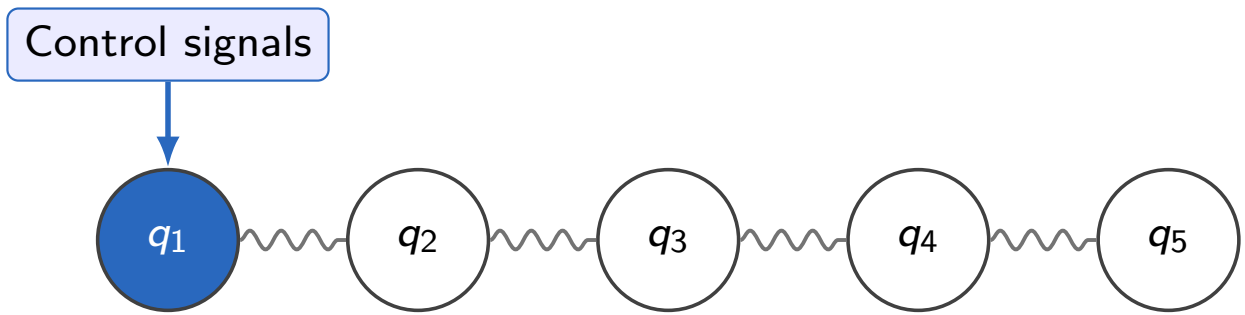
Background: quantum control

Qubits and control

A qubit stores quantum information.

Control means steering the system into a chosen state,

such as a starting state for a calculation.



Blue marks the qubits in the controlled set.

We start by controlling the qubit at one end.

A controlled qubit with exactly one uncontrolled neighbor

can force that neighbor into the controlled set.

Burgarth, Bose, Bruder & Giovannetti (2009); Burgarth et al. (2013).

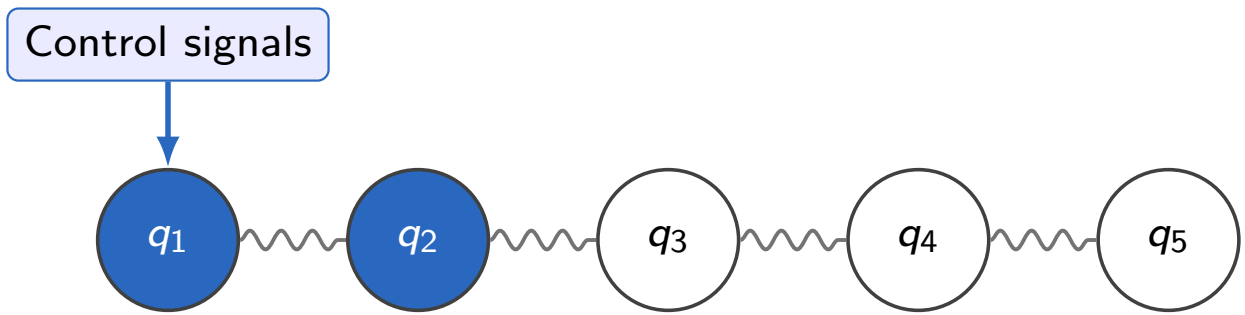
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With suitable interactions and local control,

that set can grow along the chain.

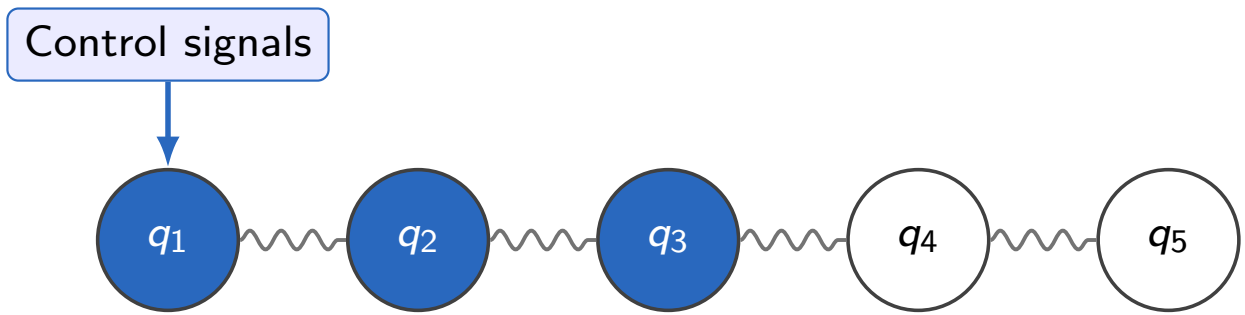
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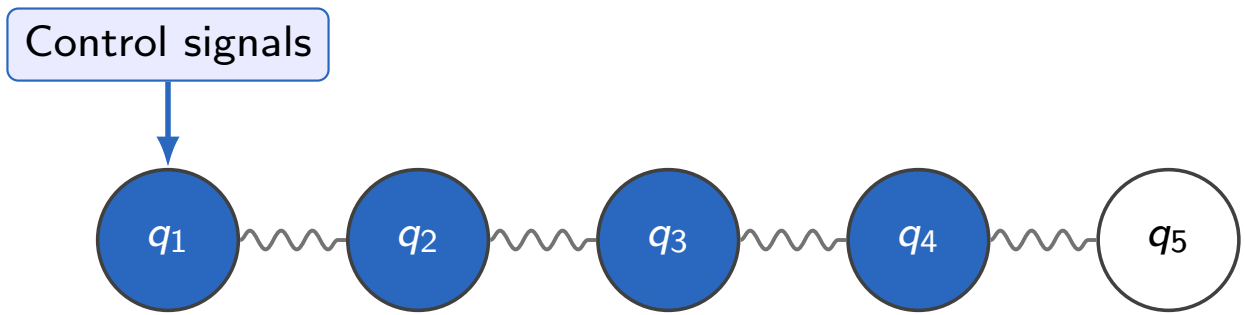
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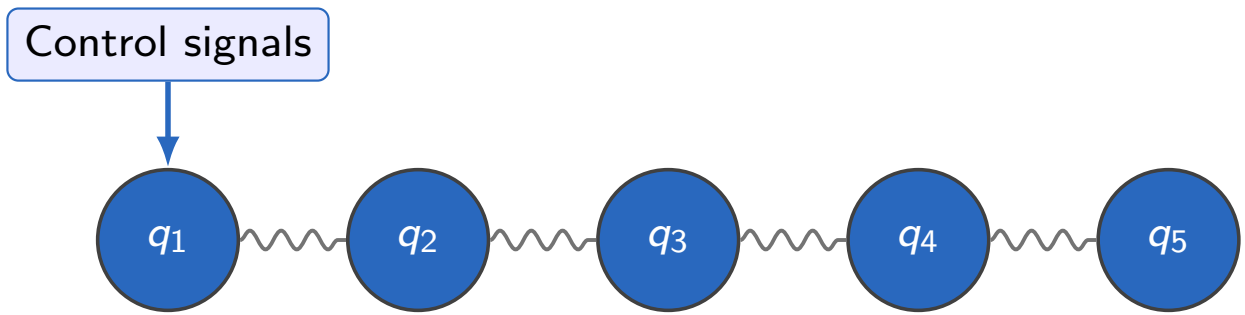
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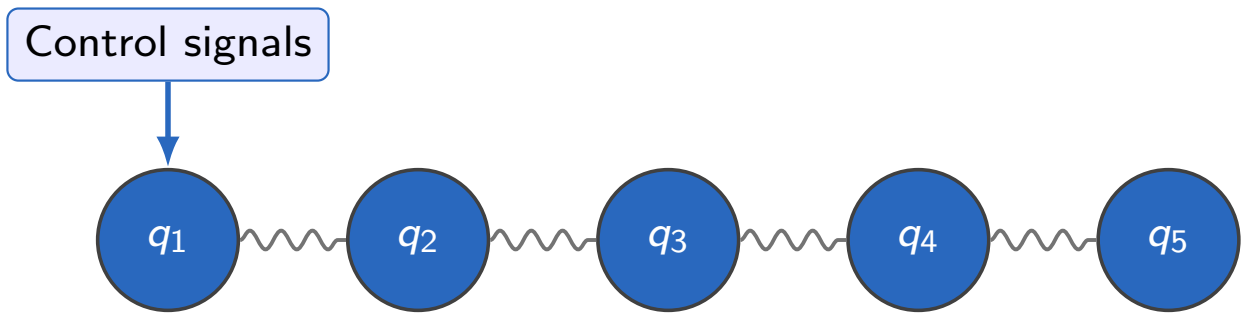
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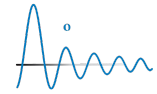


Blue marks the qubits in the controlled set.

This brings us to zero forcing, which models this spread of control in graph theory.







Our title is

Competitive Zero Forcing

on Paths and Cycles

Zero forcing

Qubits become vertices; interactions become edges.

Start with blue and white vertices.

Zero forcing

Qubits become vertices; interactions become edges.

Start with blue and white vertices.

Color-change rule

If a blue vertex has exactly one white neighbor, it forces that neighbor to become blue.

Zero forcing

Qubits become vertices; interactions become edges.

Start with blue and white vertices.

Color-change rule

If a blue vertex has exactly one white neighbor, it forces that neighbor to become blue.

Repeat the rule until no more forces are possible.

Our goal: color every vertex blue.

Blue vertices stay blue.

Standard rule: AIM Minimum Rank–Special Graphs Work Group (2008).

Zero forcing on a path

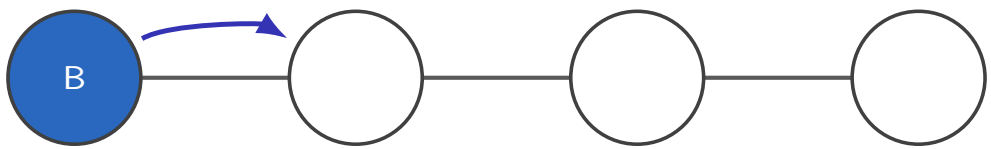
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A blue vertex with exactly one white neighbor forces that neighbor to become blue.

Zero forcing on a path

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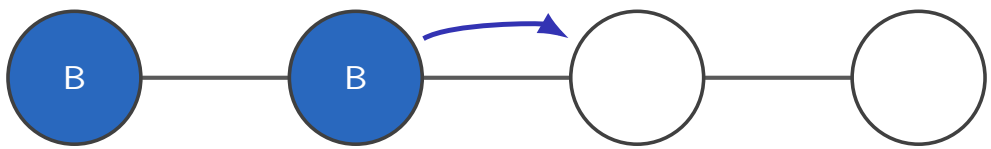
Start: the endpoint has one white neighbor.

Standard rule: AIM Minimum Rank–Special Graphs Work Group (2008).

Zero forcing on a path

Color-change rule

A blue vertex with exactly one white neighbor forces that neighbor to become blue.



Blue behind it, one white neighbor ahead:

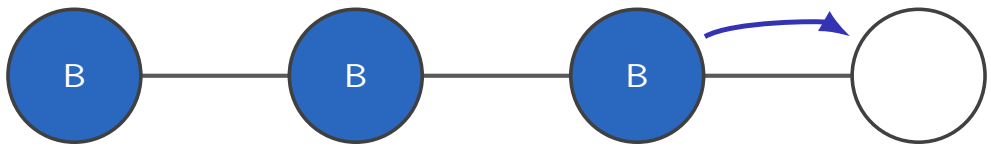
it can force again.

Standard rule: AIM Minimum Rank–Special Graphs Work Group (2008).

Zero forcing on a path

Color-change rule

A blue vertex with exactly one white neighbor forces that neighbor to become blue.



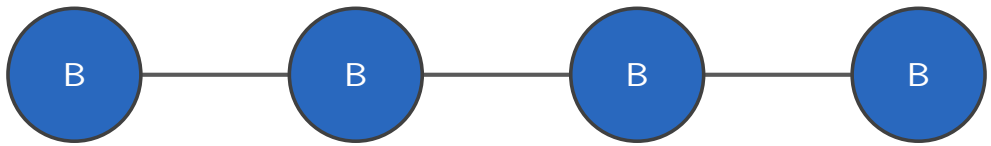
Step by step, the whole path turns blue.

Standard rule: AIM Minimum Rank–Special Graphs Work Group (2008).

Zero forcing on a path

Color-change rule

A blue vertex with exactly one white neighbor forces that neighbor to become blue.



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Zero forcing on a cycle

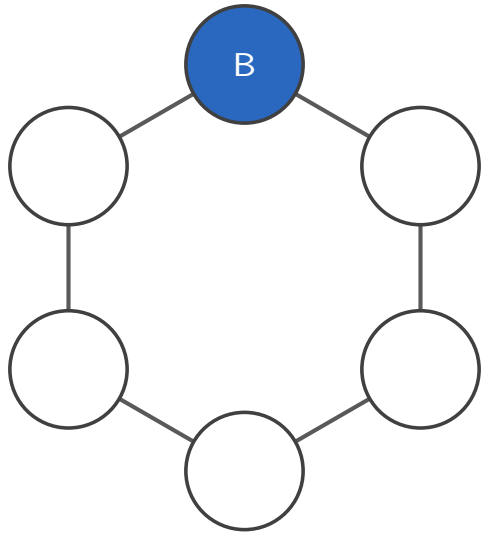
The same rule

Blue can force only when exactly one of its neighbors is white.

Zero forcing on a cycle

The same rule

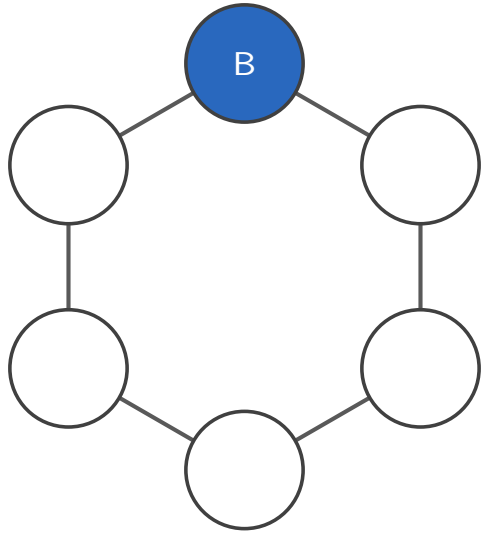
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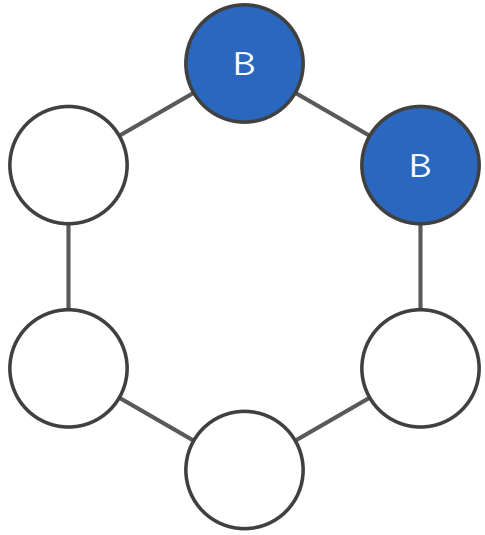


Two white neighbors: no legal force.

Zero forcing on a cycle

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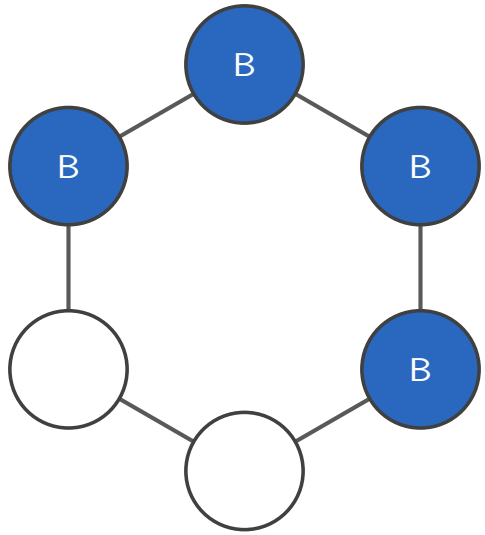
Each has one blue and one white neighbor:

both can force outward.

Zero forcing on a cycle

The same rule

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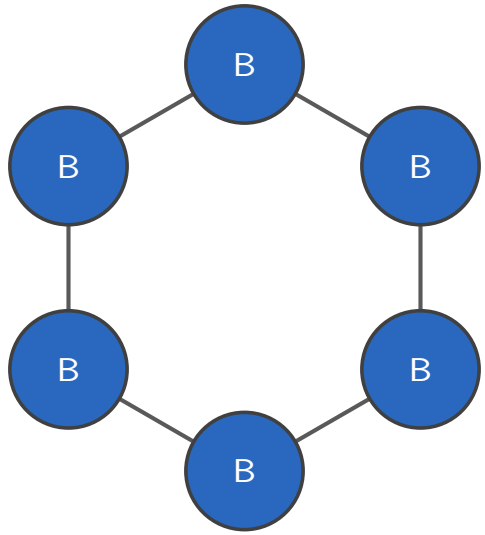


The color grows both ways until the cycle is blue.

Zero forcing on a cycle

The same rule

Blue can force only when exactly one of its neighbors is white.



An adjacent pair is enough.

Our title is

Competitive **Zero Forcing** on Paths and Cycles

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Competitive Zero Forcing

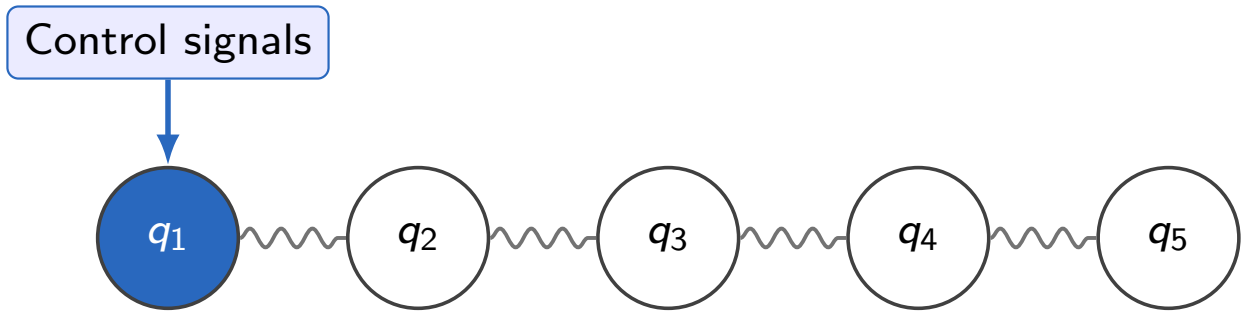
on Paths and Cycles

What changes when two colors compete?

From one set to two colors

Recall the controlled set

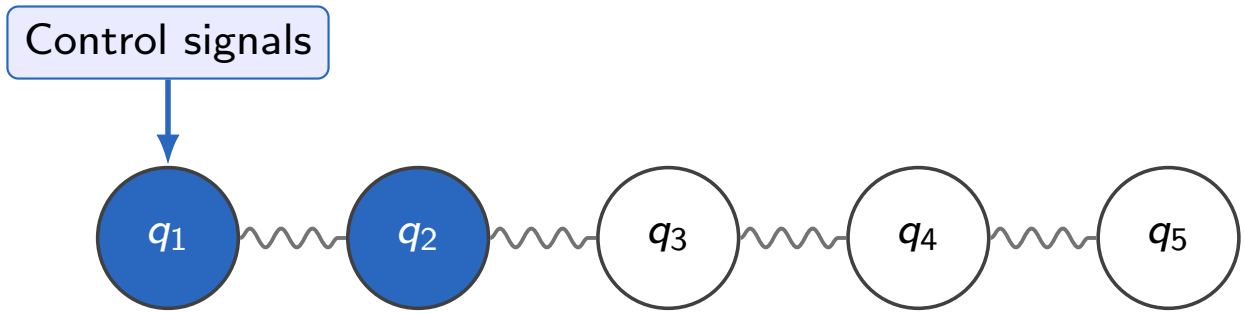
Blue shows the vertices we can control.



From one set to two colors

Recall the controlled set

When a blue vertex forces a change, we add its neighbor to that controlled set.

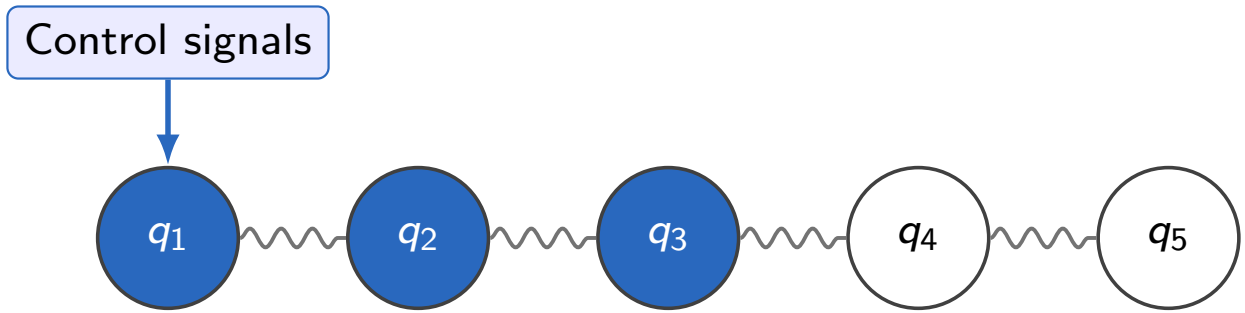


The controlled set grows by one vertex.

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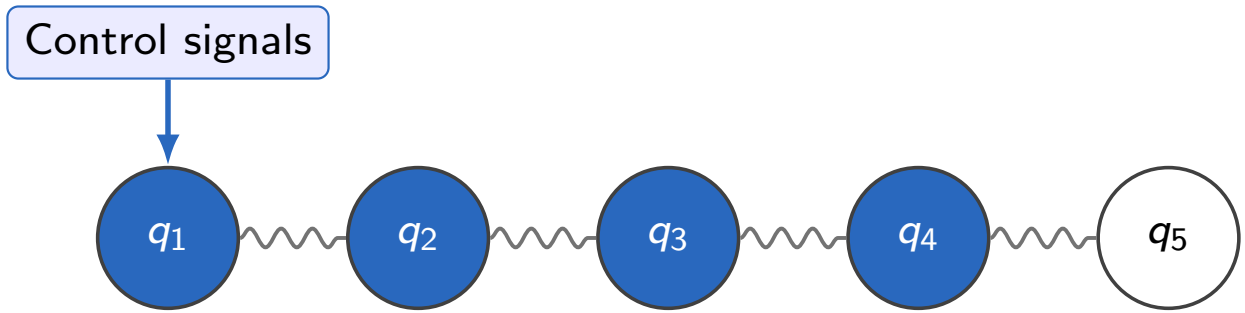


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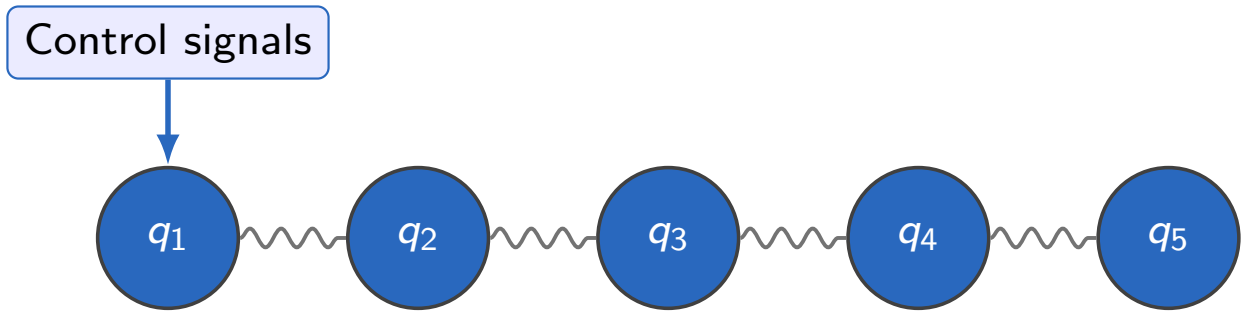


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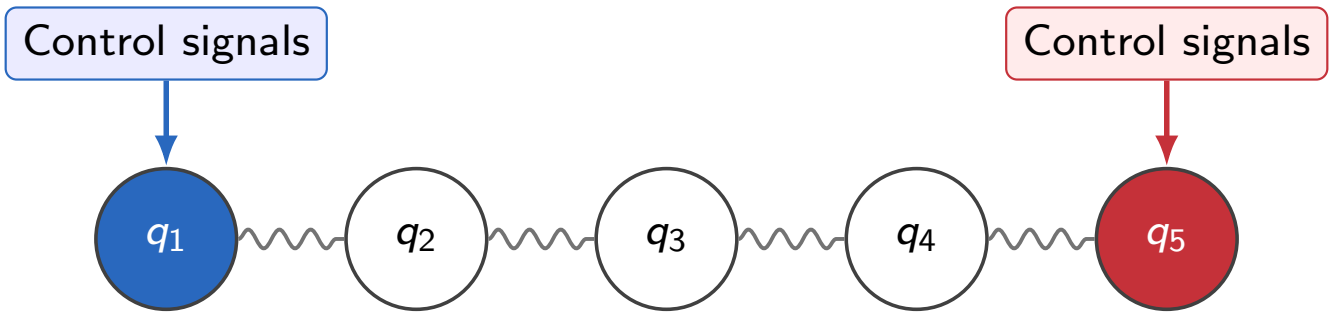


The controlled set grows by one vertex.

From one set to two colors

A new graph model

What if two starting groups spread
at the same time?

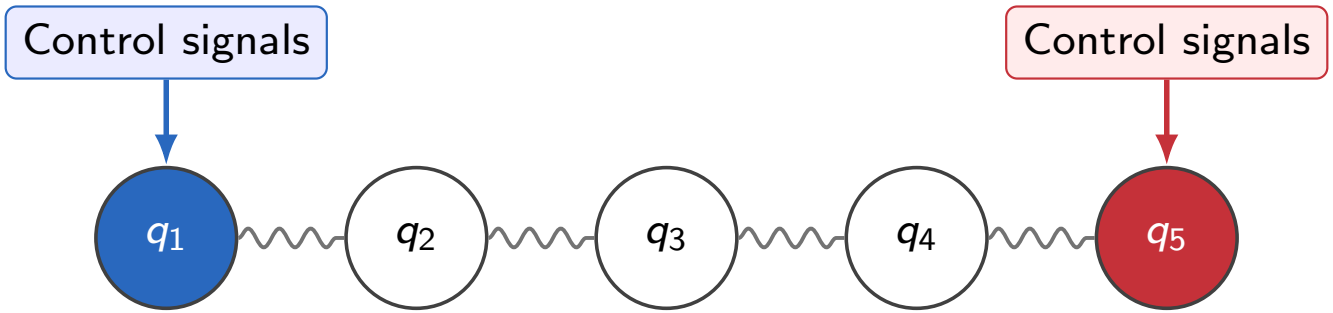


Blue and red mark the two starting groups.

From one set to two colors

A new graph model

What if two starting groups spread
at the same time?



What happens when both try to add the same neighbor?

This motivates competitive zero forcing.

Competitive zero forcing

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NSF

AIM

Force to the middle

Red forces when it has exactly one non-red neighbor.

Blue forces when it has exactly one non-blue neighbor.

Apply legal, uncanceled forces simultaneously.

Shared target

Competitive zero forcing

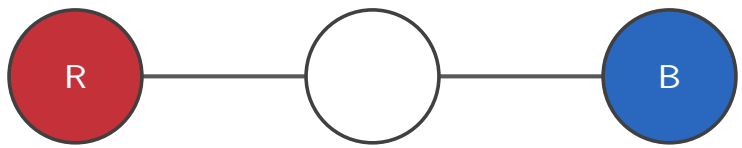
Force to the middle

Red forces when it has exactly one non-red neighbor.

Blue forces when it has exactly one non-blue neighbor.

Apply legal, uncanceled forces simultaneously.

Both colors target the middle



One red neighbor, one blue neighbor: equal support.

Competitive zero forcing

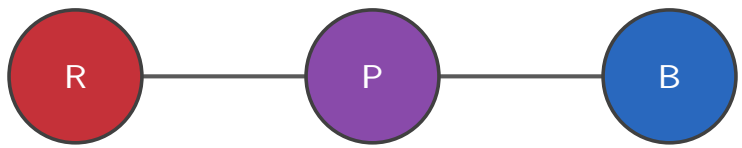
Force to the middle

Red forces when it has exactly one non-red neighbor.

Blue forces when it has exactly one non-blue neighbor.

Apply legal, uncanceled forces simultaneously.

Next round: equal support gives purple



If both colors force, count *all* red and blue neighbors.
The larger count wins; a tie gives purple.

Competitive zero forcing

Already colored target

Red has exactly one non-red neighbor: blue.

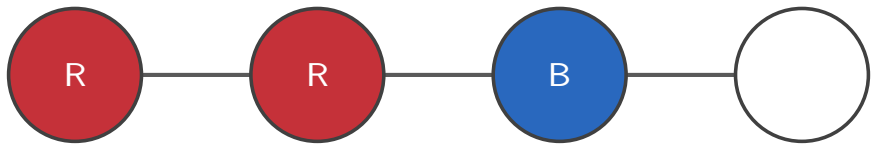
Blue has two non-blue neighbors, so it cannot force back.

Recoloring is allowed.

Already colored target

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Blue has two non-blue neighbors, so it cannot force back.



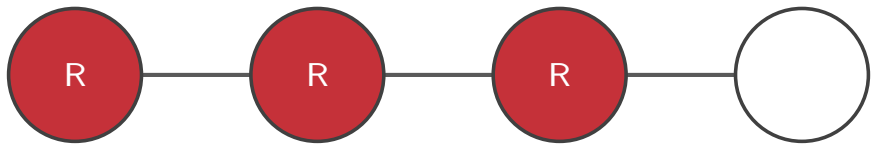
Red can force. The standalone blue vertex cannot reply.

Competitive zero forcing

Recoloring starts a chain

The new red vertex has one non-red neighbor: white.

It can force again.



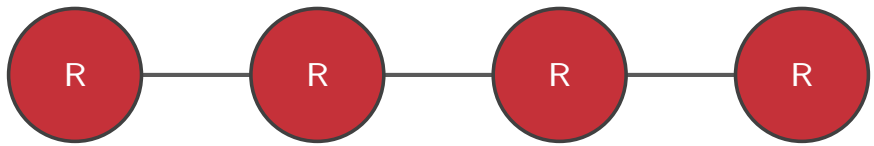
Next round: red takes over blue.

Competitive zero forcing

The chain finishes

All four vertices are red.

No white vertices remain.



One more round: red forces the last white vertex.

Competitive zero forcing

Reciprocal cancellation

If two vertices can legally force each other at round start,

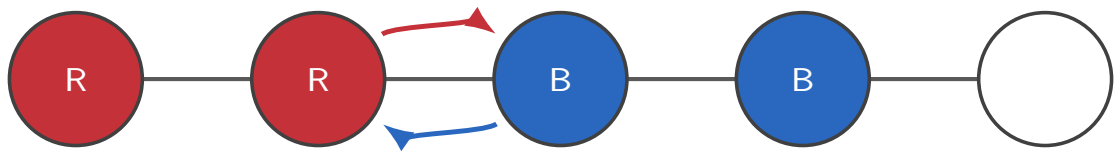
cancel both forces before updating the colors.

Competitive zero forcing

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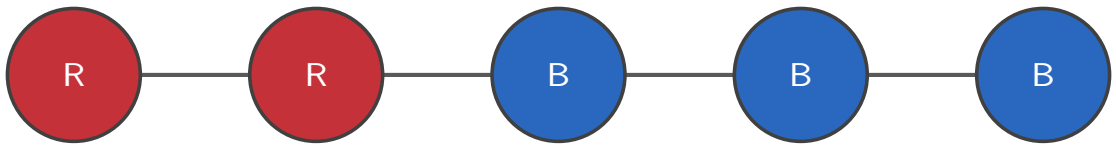
The red–blue boundary keeps its colors.

Competitive zero forcing

Reciprocal cancellation

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The boundary stays fixed; the white endpoint turns blue.

Research question

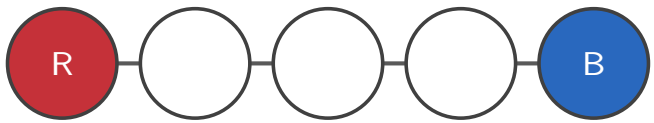
Our research question

Competitive zero forcing on paths and cycles

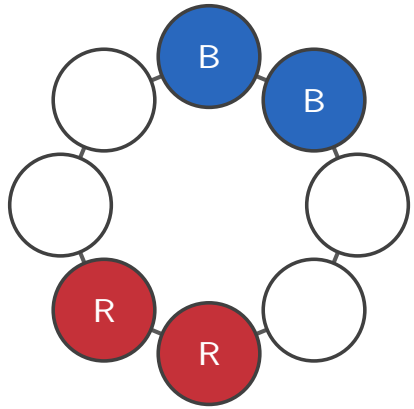
Which starting positions guarantee success?

Success: every vertex is reached, so no white vertices remain.

Path



Cycle



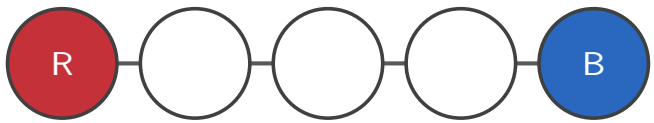
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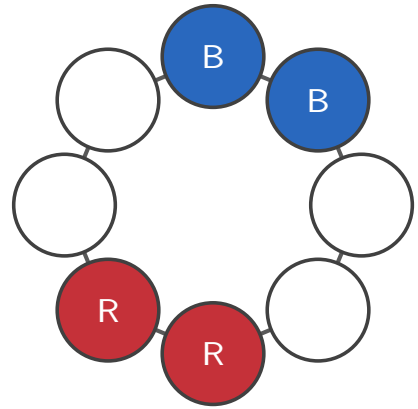
Which starting positions guarantee success?

Success: every vertex is reached, so no white vertices remain.

Path



Cycle



At that stopping point: How does initial placement affect the final outcome?

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Competitive Zero Forcing on **Paths** and Cycles

Paths: which starts succeed?

Theorem 5.2: successful starts

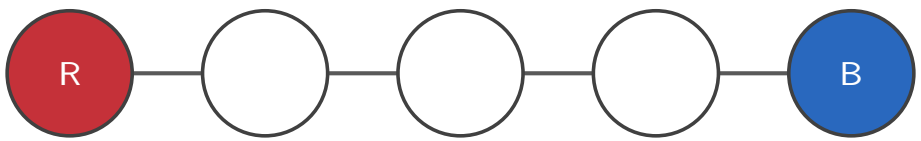
For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Start: an endpoint can force

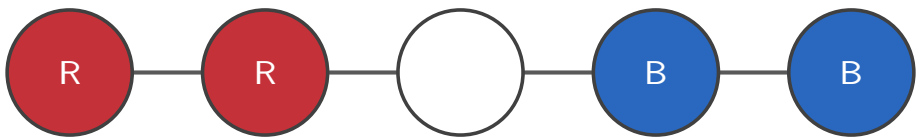


Paths: which starts succeed?

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Round 1: both ends advance

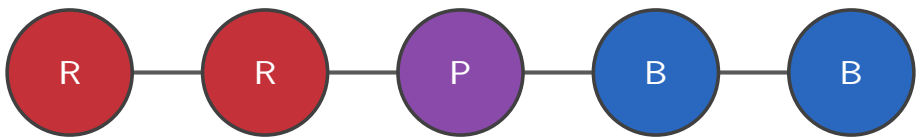


Paths: which starts succeed?

Theorem 5.2: successful starts

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Round 2: the fronts meet



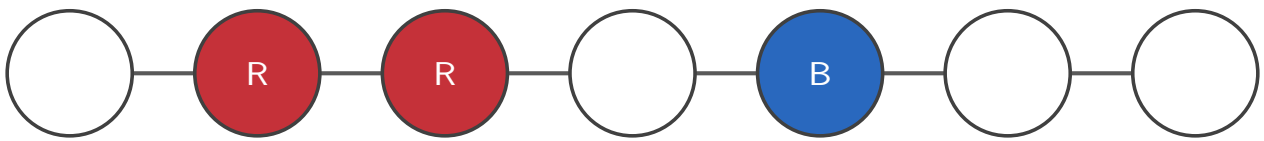
Every vertex is colored

Paths: which starts succeed?

Theorem 5.2: successful starts

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Start: a same-color adjacent pair can force

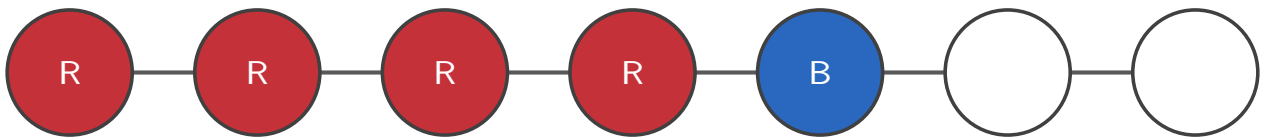


Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Round 1: red forces outward



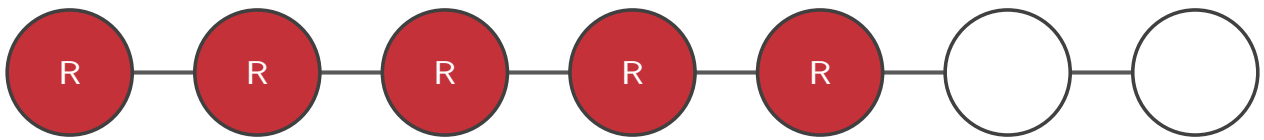
Blue has two non-blue neighbors, so it cannot force back.

Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Round 2: red takes over blue

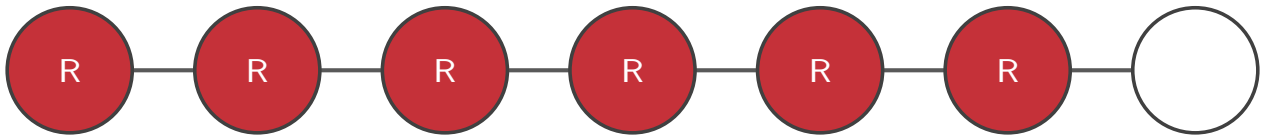


Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Round 3: red keeps spreading

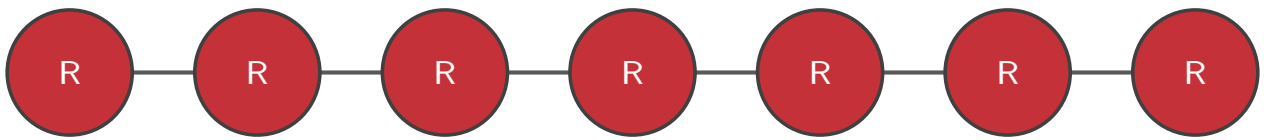


Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Round 4: no white vertices remain



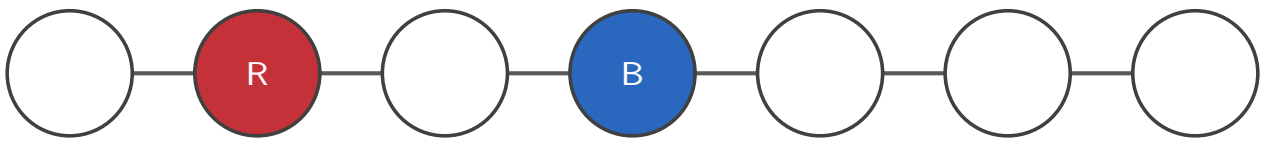
Every vertex is colored

Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Neither pattern

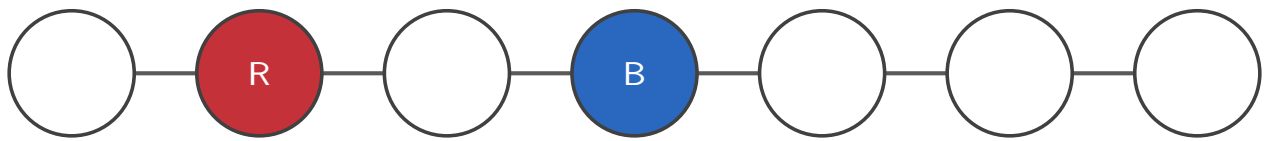


Paths: which starts succeed?

Theorem 5.2: successful starts

For $n \geq 4$, a start succeeds exactly when either color has an endpoint or a same-color adjacent pair.

Neither pattern



Each colored vertex has two neighbors outside its color.

There is no first force.

Paths: Initial placements (even)

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Proposition 5.1: final colors

One red endpoint, one blue endpoint; all other vertices white ($n \geq 2$).

$n = 2k$: k red, k blue.

$n = 2k + 1$: k of each color, one purple.

Paths: Initial placements (even)

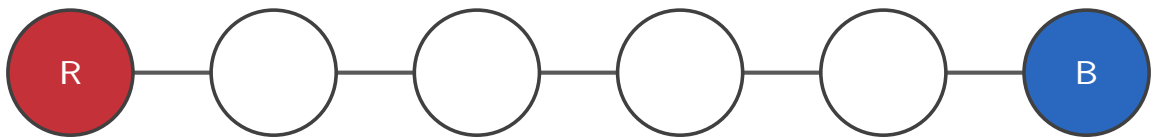
Proposition 5.1: final colors

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Six vertices: opposite-colored endpoints



Paths: Initial placements (even)

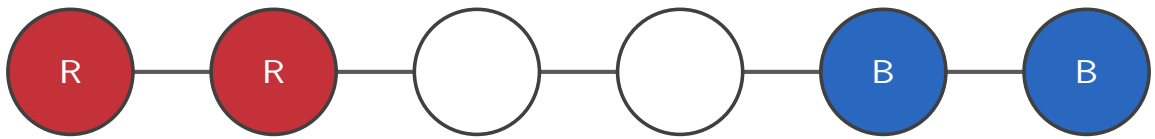
Proposition 5.1: final colors

One red endpoint, one blue endpoint; all other vertices white ($n \geq 2$).

$n = 2k$: k red, k blue.

$n = 2k + 1$: k of each color, one purple.

Both fronts advance



Paths: Initial placements (even)

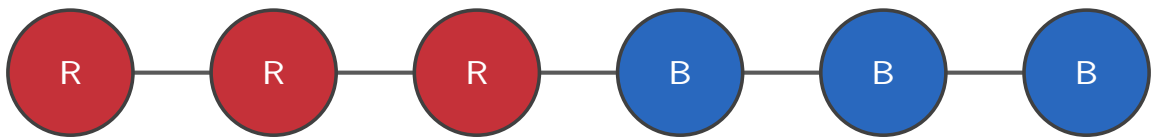
Proposition 5.1: final colors

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$n = 2k$: k red, k blue.

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The fronts meet across an edge



Paths: Initial placements (even)

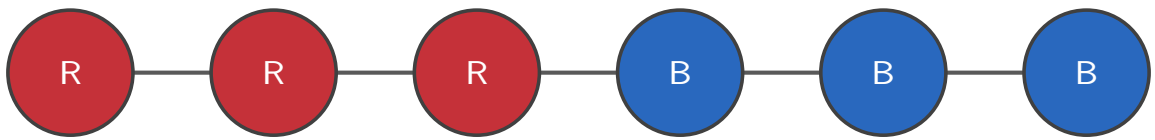
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The fronts meet across an edge



3 red, 3 blue, no purple.

Paths: Initial placements (even)

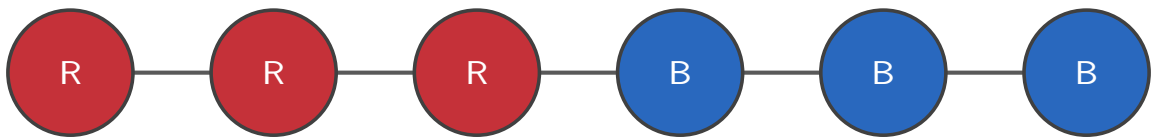
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$n = 2k$: k red, k blue.

$n = 2k + 1$: k of each color, one purple.

The fronts meet across an edge



3 red, 3 blue, no purple.

In general, $n = 2k$: k of each color.

Paths: Initial placements (odd)

Proposition 5.1: the same setup

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$n = 2k$: k red, k blue.

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Paths: Initial placements (odd)

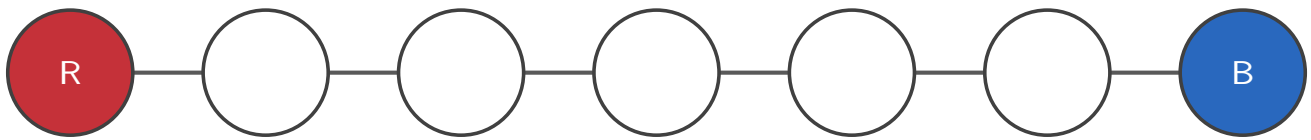
Proposition 5.1: the same setup

One red endpoint, one blue endpoint; all other vertices white ($n \geq 2$).

$n = 2k$: k red, k blue.

$n = 2k + 1$: k of each color, one purple.

Seven vertices: the same endpoint setup



Paths: Initial placements (odd)

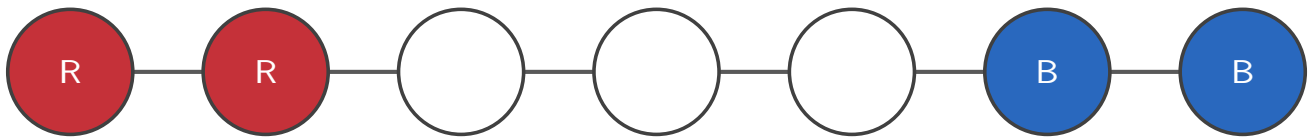
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Both fronts advance



Paths: Initial placements (odd)

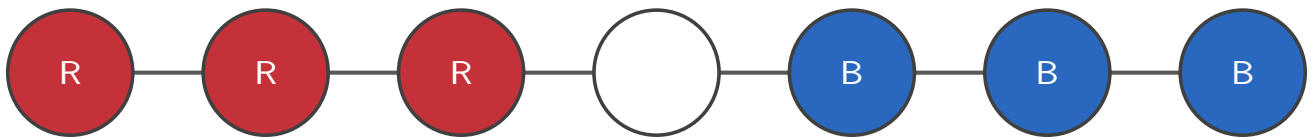
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$n = 2k$: k red, k blue.

$n = 2k + 1$: k of each color, one purple.

One white midpoint remains



Paths: Initial placements (odd)

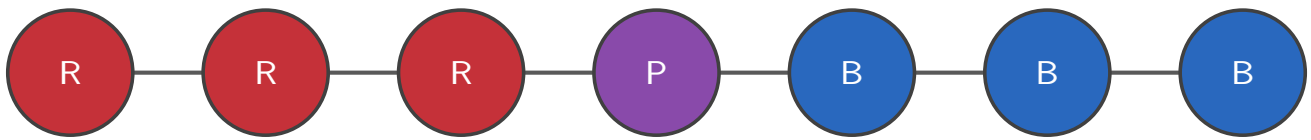
Proposition 5.1: the same setup

One red endpoint, one blue endpoint; all other vertices white ($n \geq 2$).

$n = 2k$: k red, k blue.

$n = 2k + 1$: k of each color, one purple.

Both colors reach the midpoint



Paths: Initial placements (odd)

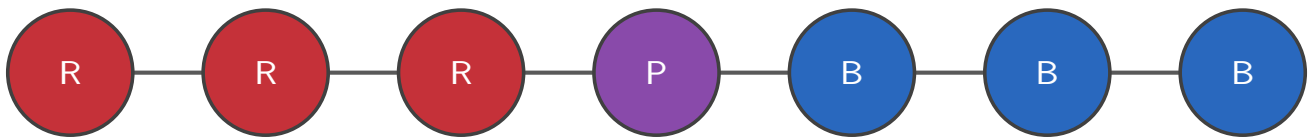
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Both colors reach the midpoint



3 red, 3 blue, one purple midpoint.

Paths: Initial placements (odd)

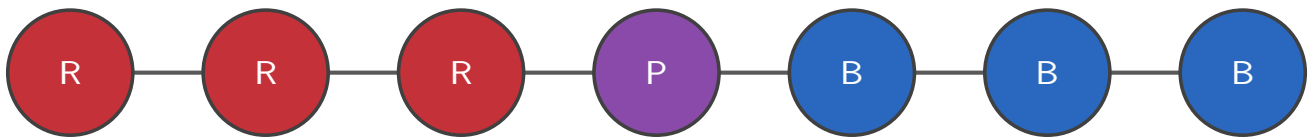
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For $n = 2k + 1$: k of each color, one purple.

Our title is

Competitive Zero Forcing on **Paths** and Cycles

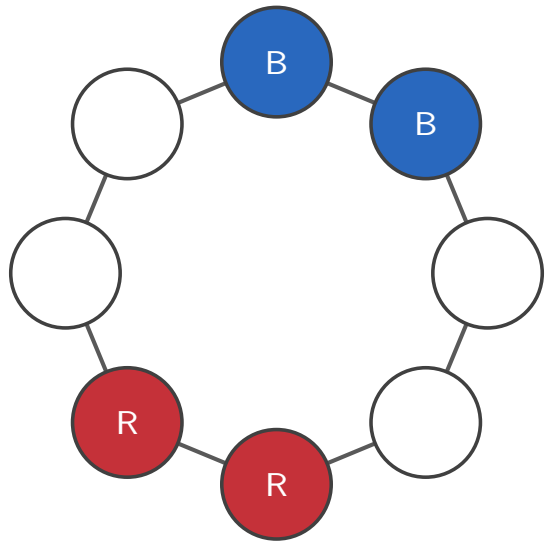
Our title is

Competitive Zero Forcing on Paths and **Cycles**

Cycles: Success

Theorem 5.5, Part 1: success

Start with exactly one red adjacent pair and one disjoint blue adjacent pair; all other vertices are white. This always succeeds.

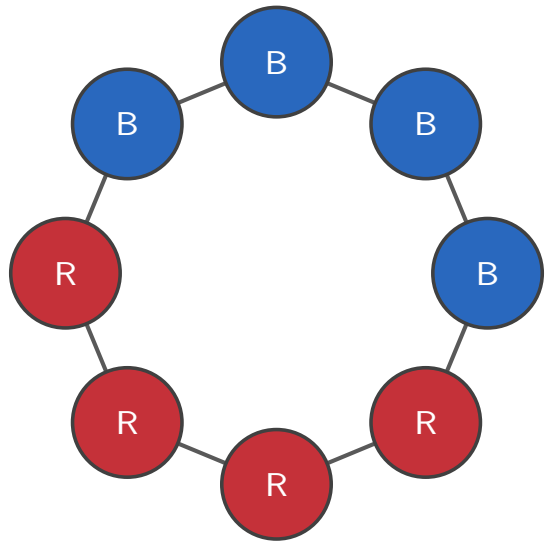


C_8

Cycles: Success

Theorem 5.5, Part 1: success

Start with exactly one red adjacent pair and one disjoint blue adjacent pair; all other vertices are white. This always succeeds.



Complete: no white vertices remain.

Cycles: Initial placements

Theorem 5.5, Part 2: final colors

Each gap adds equally many red and blue vertices.

An even gap creates no purple vertices.

An odd gap creates one purple midpoint.

Cycles: Initial placements

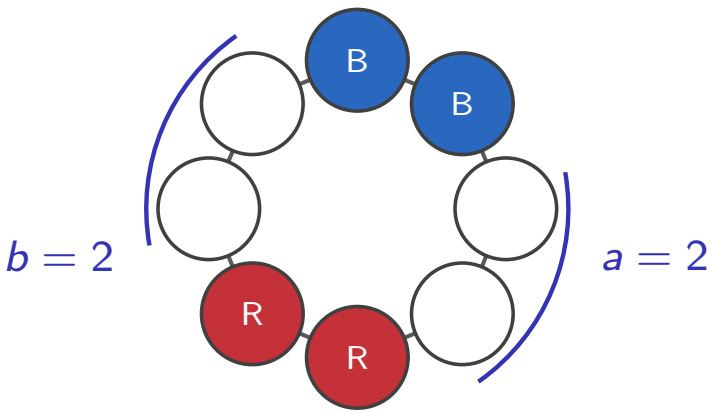
Theorem 5.5, Part 2: final colors

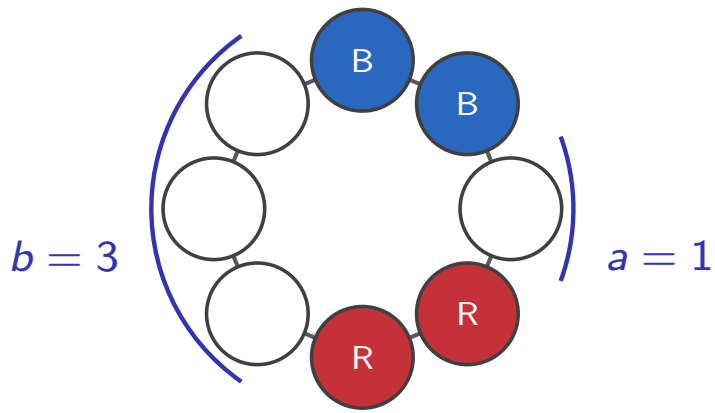
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Let a and b be the two initial white gap lengths.





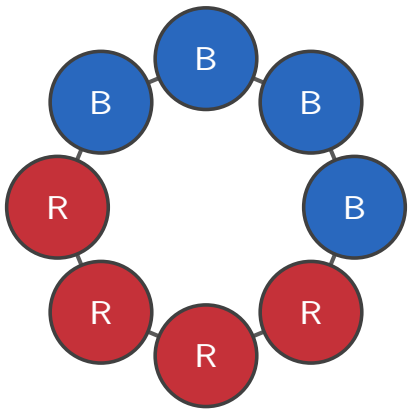
Cycles: Initial placements

Theorem 5.5, Part 2: final colors

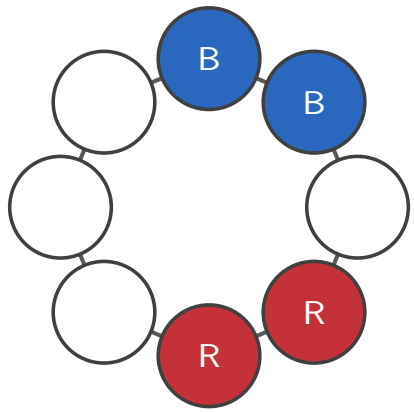
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No purple



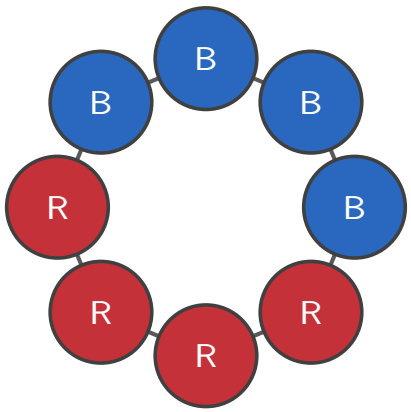
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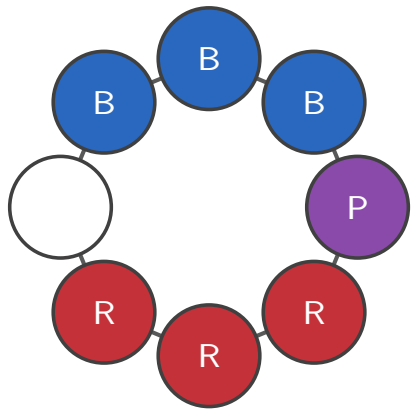
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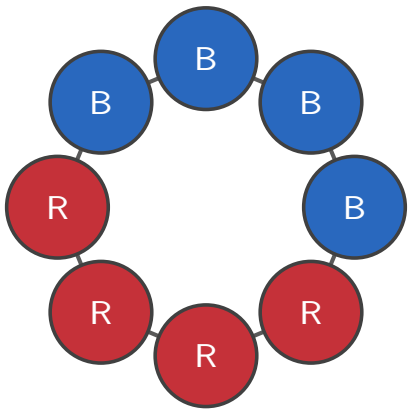
Cycles: Initial placements

Theorem 5.5, Part 2: final colors

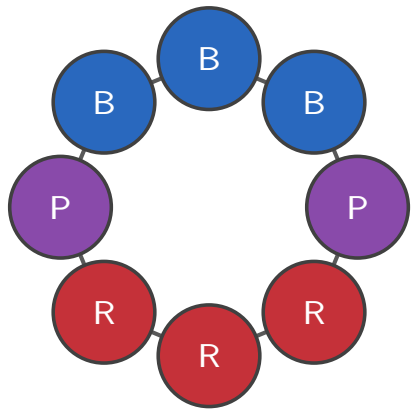
Each gap adds equally many red and blue vertices.

An even gap creates no purple vertices.

An odd gap creates one purple midpoint.



No purple



Two purple

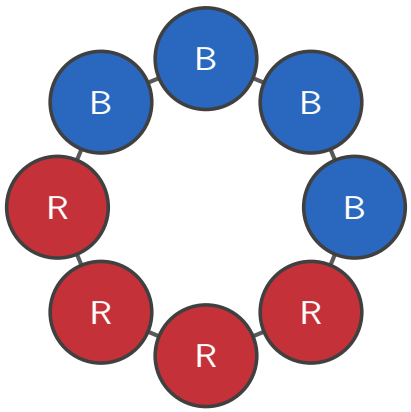
Cycles: Initial placements

Theorem 5.5, Part 2: final colors

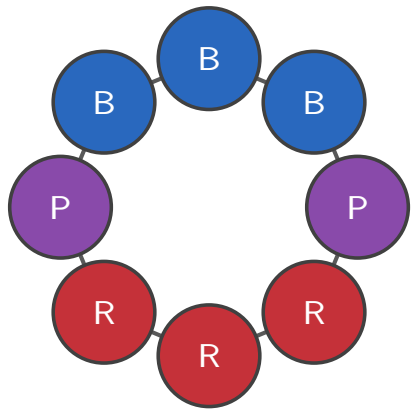
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An even gap creates no purple vertices.

An odd gap creates one purple midpoint.



No purple



Two purple

We can now answer our questions about success, initial placements, and final colors.

Conclusion

- **Paths: success.** For $n \geq 4$, an endpoint or a same-color adjacent pair in at least one color characterizes success.

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- **Cycles: initial placement.** Each odd white gap between the pairs leaves a purple midpoint.

The connection to quantum control is ongoing; our story currently ends with this graph-theoretic analysis.
Ideas or input? Please talk to me afterward or email me.

References

- Our working manuscript. *Competitive Zero Forcing*. University of St. Thomas. Work in progress.
- AIM Minimum Rank–Special Graphs Work Group. *Zero forcing sets and the minimum rank of graphs*. Linear Algebra and its Applications 428 (2008), 1628–1648.
  doi:10.1016/j.laa.2007.10.009
- D. Burgarth, S. Bose, C. Bruder, and V. Giovannetti. *Local controllability of quantum networks*. Physical Review A 79 (2009), 060305(R).
  doi:10.1103/PhysRevA.79.060305
- D. Burgarth, D. D'Alessandro, L. Hogben, S. Severini, and M. Young. *Zero forcing, linear and quantum controllability for systems evolving on networks*. IEEE Transactions on Automatic Control 58 (2013), 2349–2354.
  doi:10.1109/TAC.2013.2250075

Acknowledgments

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Dr. Mary Flagg
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Principal Investigator (PI)



Tung Vo
vo.tung@stthom.edu

National Science Foundation Grant #2331634

Support for this work

Backup: zero forcing sets

Definition

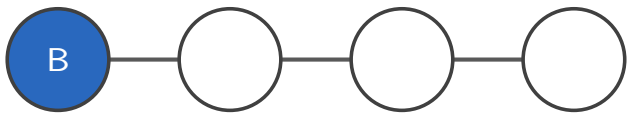
A zero forcing set is a starting set of blue vertices that eventually colors the whole graph blue.

Backup: zero forcing sets

Definition

A zero forcing set is a starting set of blue vertices that eventually colors the whole graph blue.

Path: one endpoint

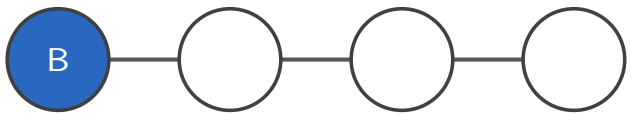


Backup: zero forcing sets

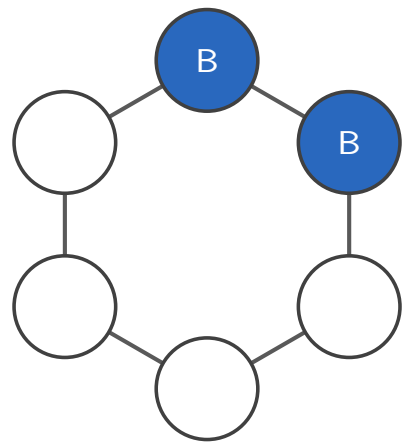
Definition

A zero forcing set is a starting set of blue vertices that eventually colors the whole graph blue.

Path: one endpoint



Cycle: an adjacent pair

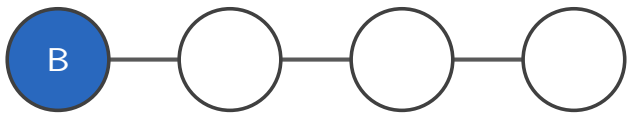


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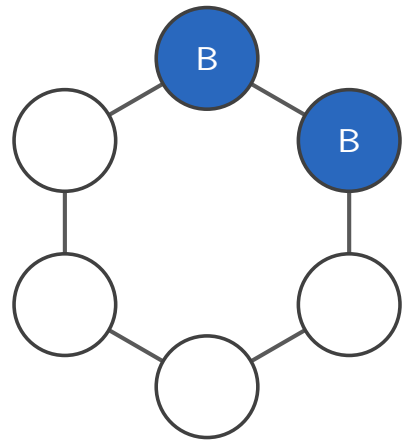
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Cycle: an adjacent pair



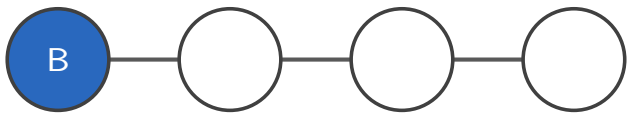
The smallest possible starting sets: $Z(P_n) = 1, \quad Z(C_n) = 2.$

Backup: zero forcing sets

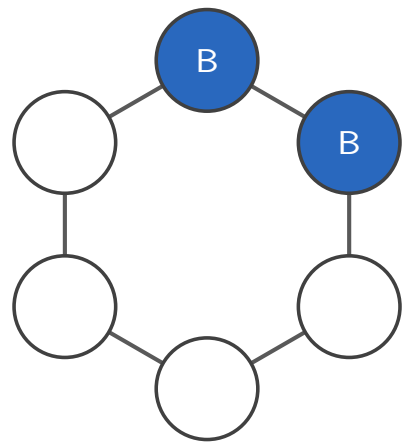
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Path: one endpoint



Cycle: an adjacent pair



The smallest possible starting sets: $Z(P_n) = 1, \quad Z(C_n) = 2.$

Starting positions matter, even with one color.

Backup: complete rules

Initially: disjoint red and blue sets, with all remaining vertices white.

Legal forces at the start of a round

Only red and blue vertices force. A red vertex u may force w if

$$N(u) \setminus R_t = \{w\} \quad \text{or} \quad N(u) \setminus (R_t \cup P_t) = \{w\}.$$

Blue uses the same conditions with B_t in place of R_t.

- Cancel $u \to w$ exactly when $w \to u$ is also legal at round start. Apply the remaining forces simultaneously.
- One forcing color: use that color. Both colors: compare all red and blue neighbors of the target. The larger count wins; a tie gives purple.
- No uncanceled force: keep the color. Targets may recolor and can force starting next round.
- Success: no white vertices, including at $t = 0$. Otherwise, a fixed or repeated coloring is failure.

Backup: path success theorem

Theorem 5.2 (our working manuscript)

Let $P_n = v_1 v_2 \cdots v_n$ be a path on $n \geq 4$ vertices, and let (R_0, B_0) be a competitive initial configuration, with all vertices outside $R_0 \cup B_0$ initially white. Then (R_0, B_0) is successful if and only if at least one of R_0 or B_0 is a standard zero forcing set of P_n.

Backup: opposite endpoints

Proposition 5.1 (our working manuscript)

Let $P_n = v_1 v_2 \cdots v_n$, with $n \geq 2$. Up to interchanging red and blue, the only competitive initial configuration in which R_0 and B_0 are disjoint minimum standard zero forcing sets is

$$R_0 = \{v_1\}, \qquad B_0 = \{v_n\},$$

with all other vertices initially white. This configuration is successful.

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$$R_0 = \{v_1\}, \qquad B_0 = \{v_n\},$$

with all other vertices initially white. This configuration is successful.

Vertices	Final coloring
Even: $n = 2k$	k red, k blue
Odd: $n = 2k + 1$	k red, k blue, one purple

Backup: cycle theorem

Theorem 5.5 (our working manuscript)

Let C_n, with $n \geq 4$, have two disjoint adjacent pairs initially colored blue and red, respectively, with all remaining vertices white. Let $a, b \geq 0$ count the white vertices in the two gaps, so that $a + b = n - 4$.

Under simultaneous competitive zero forcing with reciprocal forces canceled, this configuration is successful. Each gap of length g contributes $\lfloor g/2 \rfloor$ blue vertices and $\lfloor g/2 \rfloor$ red vertices, together with one purple vertex at its midpoint if g is odd.

Backup: cycle color counts

Theorem 5.5: final color counts

Vertices and gaps	Red	Blue	Purple
Odd: $n = 2m + 1$	m	m	1
Even: $n = 2m$, both gaps even	m	m	0
Even: $n = 2m$, both gaps odd	$m - 1$	$m - 1$	2

Backup: cycle color counts

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For C_8: even gaps give no purple; odd gaps give two purple.